Saul Harwin

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Physics graduate (2:1 BSc) with a strong foundation in analytical thinking, quantitative methods, and programming, developed through academic and self-directed projects. Experience includes building simulations, working with data, and applying statistical and computational techniques to solve complex problems. Confident translating theoretical concepts into structured, efficient code.

Alongside technical work, developed communication and teamwork skills through volunteering, including teaching and coordinating activities with children. Seeking a junior software engineering role to apply and further develop skills in data science, machine learning, and analytics within a collaborative development environment.

Education

University of Sussex
2022 – 2025
BSc Physics (2:1)

Key modules include:

Mathematical Methods
Scientific Computing
Quantum Mechanics
Electrodynamics

Uckfield College
2020 – 2022

A Level Results:

Physics **(A)**
Mathematical Science **(A)**
Computer Science **(A)**

Projects – Additional Detail can be found on my website.

Weak Gravitational Lensing Analysis – Final Year Project, University of Sussex

- Implemented a weak gravitational lensing model using the Singular Isothermal Sphere (SIS) framework, translating theoretical mass distributions into observable shear signals
- Processed and validated 180,000 galaxy ellipticity measurements, extracting tangential shear profiles while mitigating noise and statistical bias in the dataset
- Designed a statistical fitting pipeline incorporating error propagation and residual analysis, achieving a robust SIS model fit ($\chi^2 = 0.86$)
- Performed sensitivity analysis on noise models, quantifying their impact on mass estimation accuracy and identifying SIS model breakdown in high-density core regions

Tools: Python (NumPy, SciPy, Matplotlib), statistical modeling, astrophysical data analysis & visualization.

Monte Carlo Simulations Dissertation – University of Sussex

- Implemented Monte Carlo and MCMC algorithms in R to solve problems in stochastic processes and numerical analysis, focusing on probabilistic modelling and simulation accuracy.
- Developed random sampling pipelines (inverse transform), validating distributions via Kolmogorov-Smirnov tests and applying importance sampling for efficient expectation estimation.
- Implemented MCMC random walk simulations to analyse Markov chain properties, debugging convergence behaviour and verifying ergodicity and mixing characteristics.
- Approximated solutions to Laplace and Poisson PDEs using Monte Carlo random walk methods, evaluating convergence rates and error-computation trade-offs.
- Ensured reproducibility of stochastic simulations through controlled random seeding and statistical consistency checks.

Tools: R, MCMC, Importance Sampling, Probability Theory, Error Metrics, Statistical Validation

F1 Season Driver Analytics Project – Personal Project

- Developed a data analytics platform to explore and visualise Formula 1 driver and season performance using historical race data.
- Designed interactive dashboards and visualisations to identify performance trends, compare drivers, and analyse championship outcomes.
- Collected, cleaned, and transformed large datasets, applying statistical analysis techniques to generate meaningful insights.
- Built and deployed a responsive web application, demonstrating end-to-end software development and data engineering skills.

Key Skills: Data analysis, SQL, Python, data visualisation, problem-solving, data cleaning.

Flocking Bird Simulation – Unity project using Boids algorithm

- Implemented a real-time agent-based simulation in Unity (C#) using Boids rules, modelling emergent flocking behaviour through local interaction dynamics.
- Calibrated simulation parameters through iterative testing, applying vector-based kinematics and Unity physics to stabilise agent motion and achieve realistic group dynamics.
- Designed and debugged real-time update loops and agent interactions, ensuring stable performance and consistent behaviour in a non-deterministic simulation environment.

Tools: Unity, C#, Vector Math, Game Physics.

For additional projects and to see detailed descriptions of my projects please visit my website saulharwin.com

Other Experience

2019 – Present: Volunteer: Pevensey Bay Sailing Club (Assistant Instructor)

- Planned and delivered structured training sessions for groups of up to 5 children (ages 7–16), coordinating with a team to ensure safe and effective instruction.
- Adapted instructional methods to suit varying ages and learning styles, ensuring clear communication of practical and safety-critical information.
- Applied safeguarding protocols and maintained continuous situational awareness to ensure participant safety in a dynamic, risk-sensitive environment.

Key Skills: Leadership, instructional clarity, adaptability, responsibility.

June – August 2024: Clover Construction Ltd (Labourer)

- Demonstrated strong time management skills by efficiently completing tasks within tight deadlines on a busy construction site.
- Gained practical problem solving experience by tackling onsite challenges in changeable conditions.
- Trained in site safety awareness, including duty of care to colleagues, clients, and the public, and preventing environmental harm.
- Executed tasks in accordance with detailed technical plans, adapting to evolving requirements while maintaining accuracy and consistency.

Key Skills: Problem-solving, team collaboration, task prioritization, workplace safety culture.

November – December 2023: Catsfield Christmas Tree Farm (Gardener)

- Quickly understanding customer's problems and acting to resolve them in a dynamic environment.
- Manager vehicle movements on site and in the car park during busy periods, ensuring site safety and that disruption on the public road was minimised.
- Carrying out agreed processes were adhered to, to maximise efficiency and ensure the job is performed correctly first time.

Key Skills: Operational efficiency, process following, customer service, situational awareness.

July - September 2022: CURIOUS TECH LTD (iPhone Technician)

- Diagnosed and repaired iPhones, delivering high-quality refurbishments to meet performance and cosmetic standards.
- Interfaced with customers to assess device issues, offering tailored technical advice and support.
- Entered and updated device information in the backend system to maintain accurate inventory and tracking records, ensuring an efficient and timely refurbishment proccess.
- Carried out strict data privacy by securely wiping devices and handling customer data in compliance with company protocols.

Key Skills: Technical diagnostics, user support, data integrity, secure handling of sensitive information.

September 2021 – April 2022: La Stalla Pizzeria e Ristorante (Assistant Cook)

- Ensured customer allergies are understood and hygiene standards are maintained.
- Improved my communication skills in a fast passed work environment.
- Developed my capabilities to make quick decisions under high stress situations.

Key Skills: Multitasking, communication, working under pressure, interpersonal skills.

Awards and Certificates

- **Machine Learning Course (Standford Online, Deeplearning.AI):** Learning foundationary machine
- learning skills and procedures.
- **Citb Health and Safety Awareness Certiticate:** Developing my awareness of health and safety in the construction industy.
- **RYA First Aid Course:** I have completed a first aid course that allows you to learn to become an Assistant Sailing Instructor.
- **Assistant Sailing Instructor:** To become an instructor at Pevensey Bay Sailing Club I have passed a course.
- **RYA Dinghy Level 4:** The Highest dinghy sailing certificate demonstrating advance sailing ability in varied conditions.
- **RYA Powerboat Level 2:** Equiping me with the skills and knowledge needed to safely operate a powerboat.